

① $\frac{\partial Y_{t+2}}{\partial \theta_t} = ?$

Y_t : Stationary.

θ_t

"

Data.

Wold form

(representation)

$Y_t = \underbrace{\phi_0 \theta_t}_{\text{Deterministic part}} + \underbrace{\phi_1 \epsilon_{t-1} + \phi_2 \epsilon_{t-2} + \phi_3 \epsilon_{t-3}}_{\text{Stochastic part}}$

Deterministic part

Stochastic part

Financial Econometrics:

AutoRegressive MovingAverage (ARMA) models

Approximation

Z-R
($\frac{1}{\phi_0} \theta_t - \theta_t$)

② $\theta_t | \bar{\theta}_t$

θ_t Data

θ_t Data

Sora Chon

Inha University

(Stationary)

Any time-series Data $\Rightarrow$ Wold form

$\Rightarrow \underline{\underline{\infty}}$ parameters

$\Rightarrow$ approximation

"ARMA"

Motivation

MA Model

AR Model

Wold representation and impulse response analysis

Prediction

Generalization

Outline

Motivation

MA Model

AR Model

Wold representation and impulse response analysis

Prediction

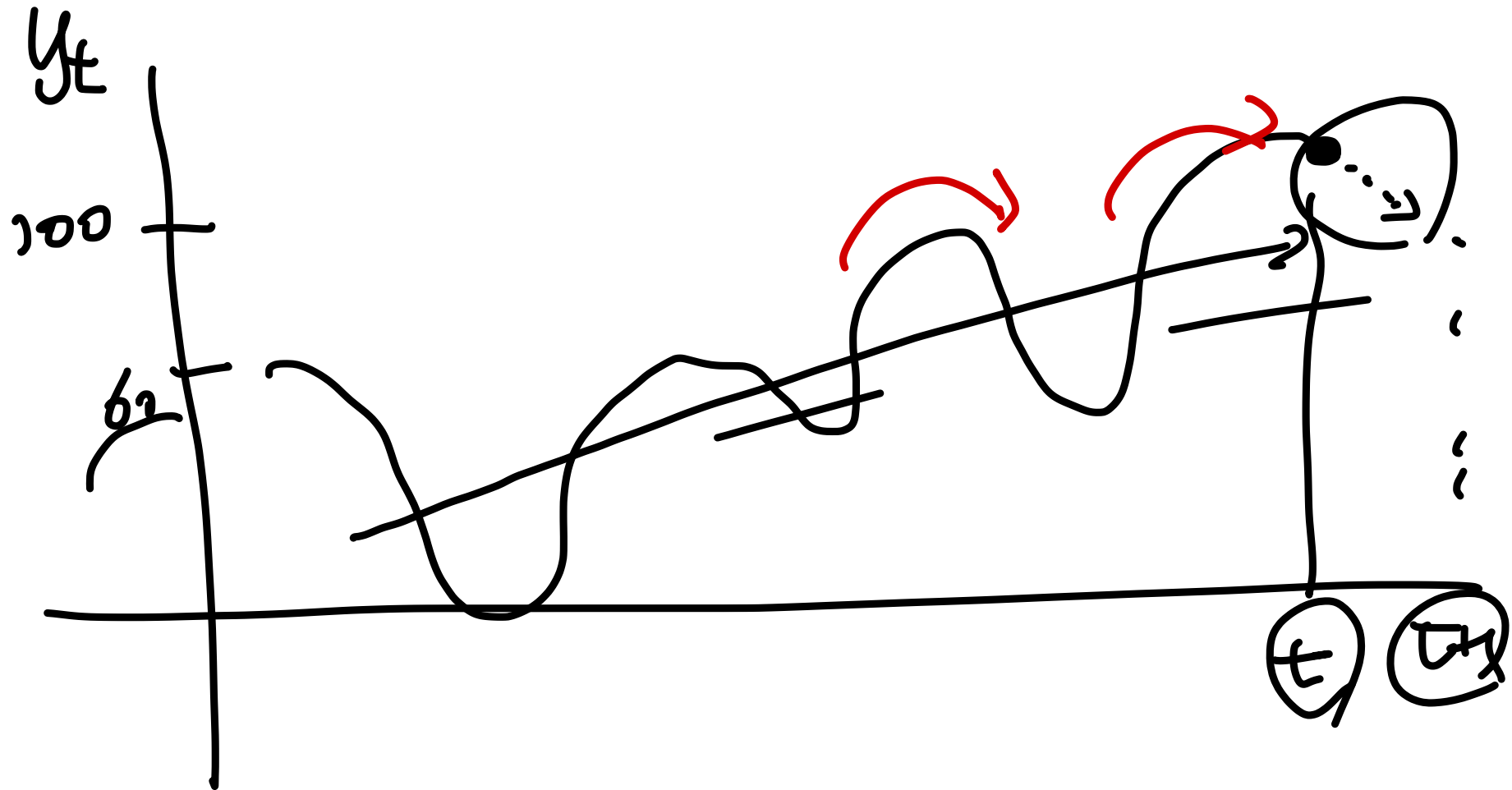
Generalization

Motivation

- ▶ What is the simplest linear univariate model for time series data?
- ▶ It is the AutoRegressive MovingAverage (ARMA) model.
- ▶ What does it do? The model captures the past data (repeated) patterns.

Motivation

- Why capturing the past data pattern is helpful for forecasting?



Motivation

- ▶ Other uses?
- ▶ Checking stationarity
- ▶ Forecasting
- ▶ Causal inference (not good though...)

Time series model

- ▶ Def: A time series model is a probability model to describe the behavior of a stochastic process $\{Y_t\}$.
- ▶ Note: Typically, a time series model is a simple probability model that describes the time dependence in the stochastic process $\{Y_t\}$.

Time series model

AR: Auto - Regressive
MA: Moving - Average.

ARMA(p, q) model (mean-adjusted form):

$$Y_t - \mu = \phi_1 (Y_{t-1} - \mu) + \cdots + \phi_p (Y_{t-p} - \mu) \\ + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q} \\ \varepsilon_t \sim N(0, \sigma^2)$$

or

white noise

~~MA~~

ARMA(p, q) model (regression form):

$$Y_t = c + \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} \\ + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q} \\ \varepsilon_t \sim N(0, \sigma^2)$$

Time series model

AR(2) model (regression form):

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t$$
$$\varepsilon_t \sim N(0, \sigma^2)$$

MA(2) model (regression form):

$$Y_t = c + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$$
$$\varepsilon_t \sim N(0, \sigma^2)$$

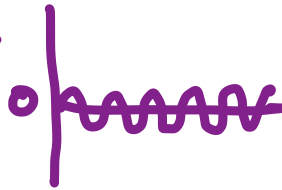
ARMA(1, 2) model (regression form):

$$Y_t = c + \phi_1 Y_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$$
$$\varepsilon_t \sim N(0, \sigma^2)$$

Motivation

$$y_t = \mu + \phi_1 y_{t-1} + \epsilon_t$$

$\epsilon_t \sim N(0, \sigma^2)$

e.g. 

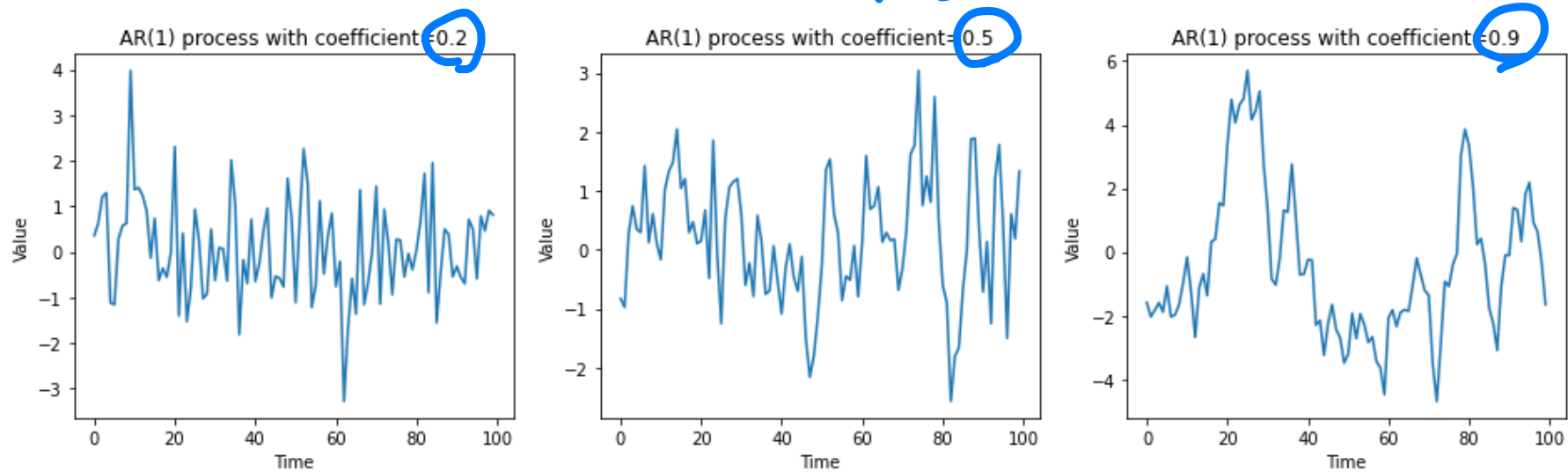


Figure: AR(1) processes

- AR(1) coefficient determines persistence.

Motivation

$$y_t = c + \epsilon_t + \theta_1 \epsilon_{t-1}$$

MA(1) $\epsilon_t \sim N(0, \sigma^2)$

$$\phi_1 = 0.8$$

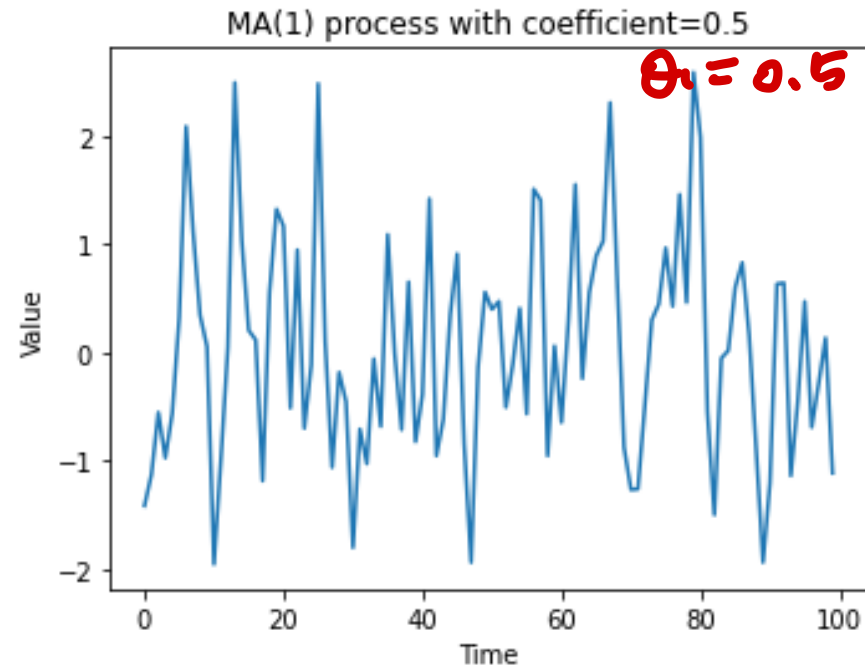
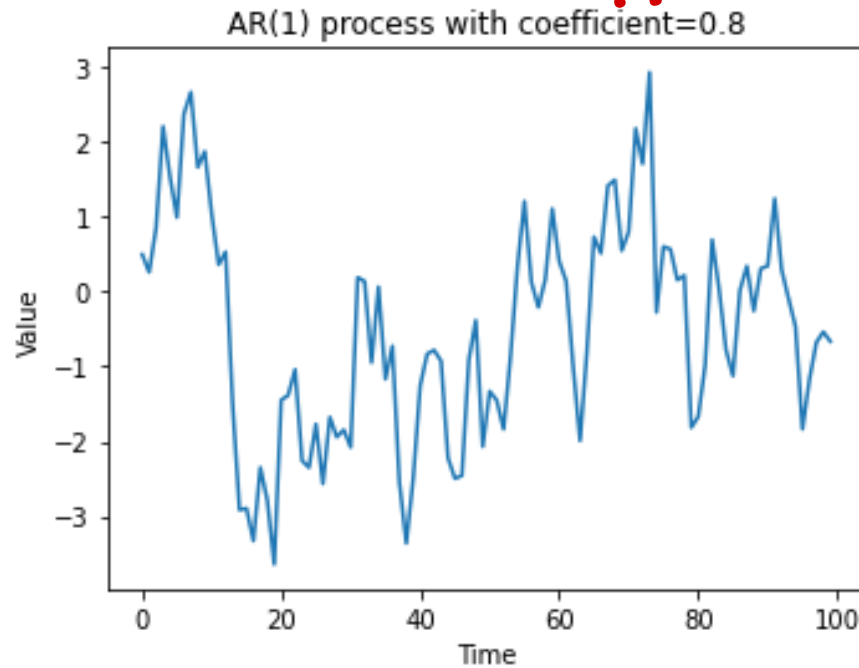


Figure: AR(1) and MA(1) processes

- ▶ AR(1) process typically show more persistent movements than MA(1) process.

Motivation

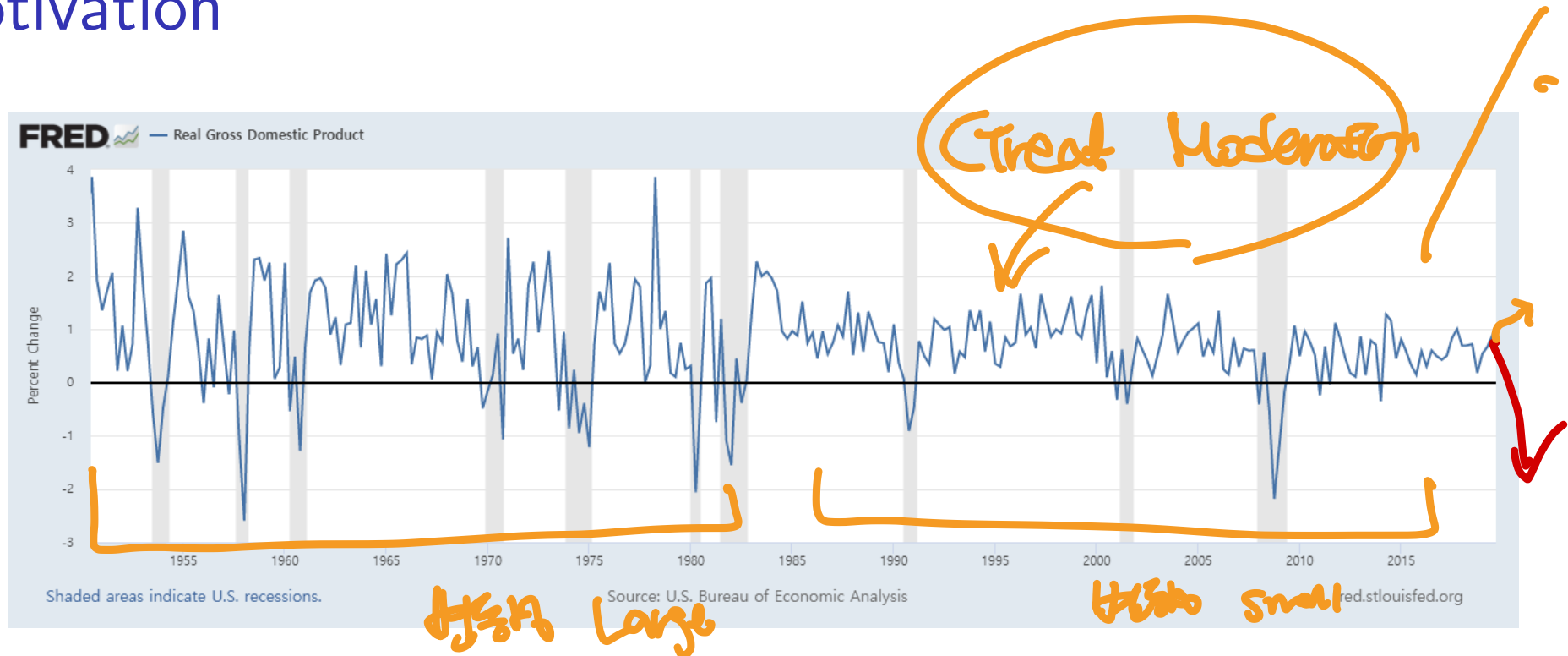


Figure: U.S. real GDP growth rate (quarterly)

- ▶ Which ARMA model should be used to fit the time series data?
- ▶ We need to know the properties of each ARMA model to find the best ARMA model. That means choosing the orders of AR and MA terms.

Outline

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MA Model

AR Model

Wold representation and impulse response analysis

Prediction

Generalization

Moving Average (MA) Processes

$\begin{pmatrix} \text{기분} \\ \text{기분} \end{pmatrix} \leftarrow \begin{pmatrix} \text{변수} \\ \hat{\beta} ? \\ E(\hat{\beta}), \text{var}(\hat{\beta}) ? \end{pmatrix} \quad \text{e.g.} \quad \underline{Y_i} = \beta \underline{x_i} + e_i$

- Idea: Create a stochastic process that only exhibits one period linear time dependence

84
 ↓
Estimation
 ↓
7-R

MA(1) Model

$$Y_t = \mu + \varepsilon_t + \theta \varepsilon_{t-1}, \quad -\infty < \theta < \infty$$

$$\varepsilon_t \sim \text{iid } N(0, \sigma_\varepsilon^2)$$

θ determines the magnitude of time dependence

추측이 많을수록
 반응이 아주 작다.

$$\frac{\partial Y_{t+1}}{\partial \varepsilon_t} = ?$$

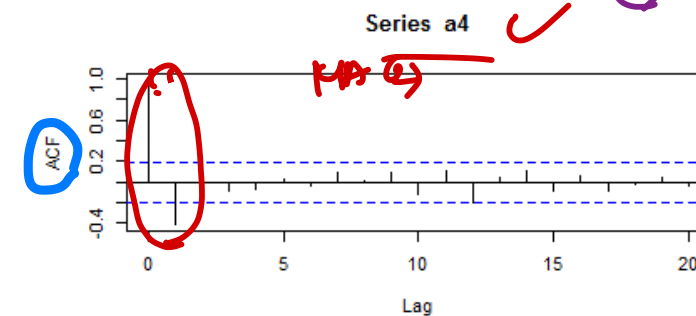
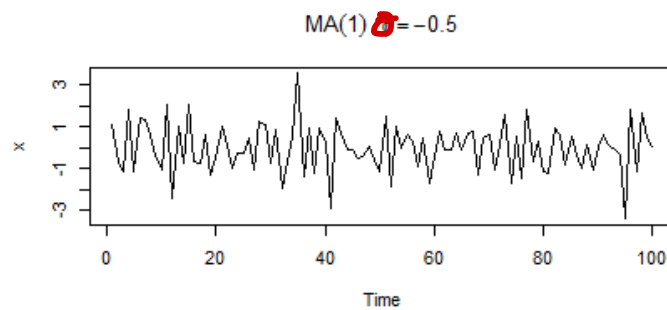
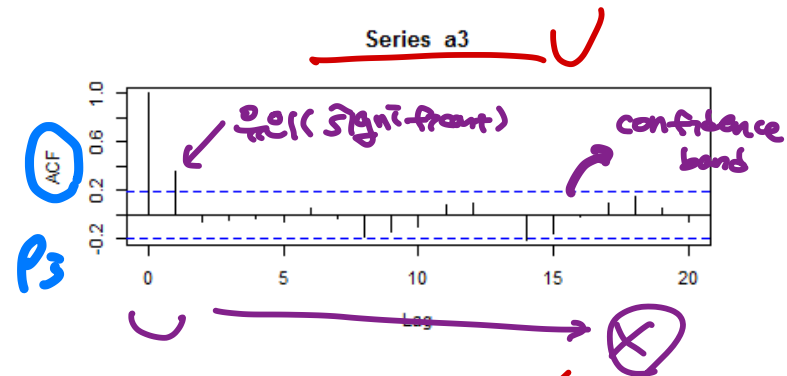
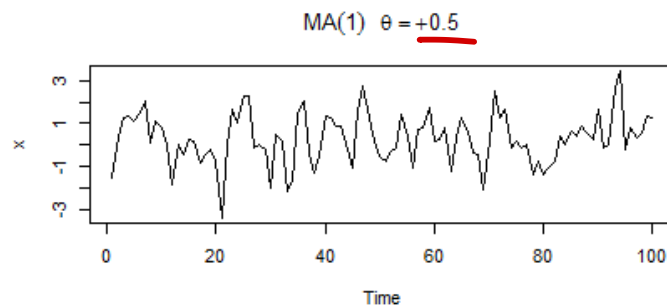
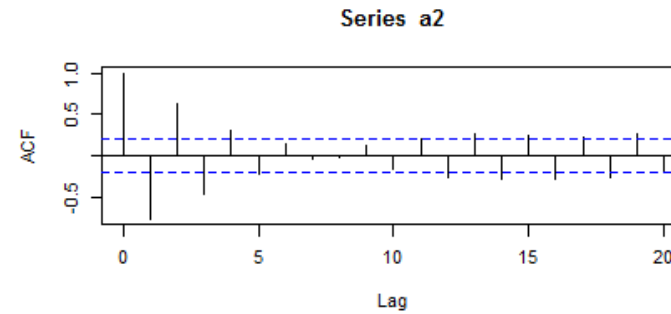
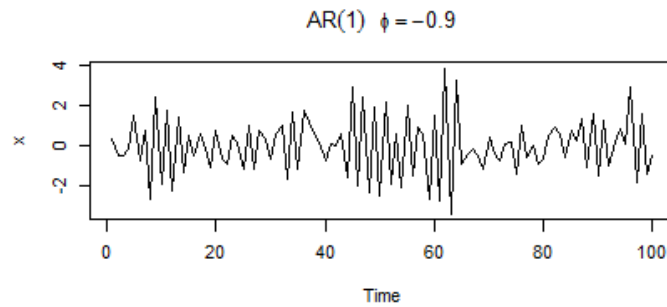
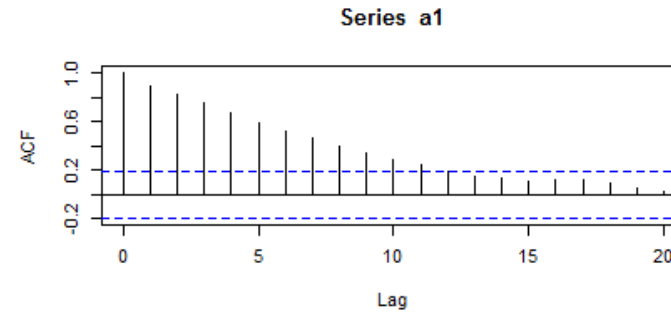
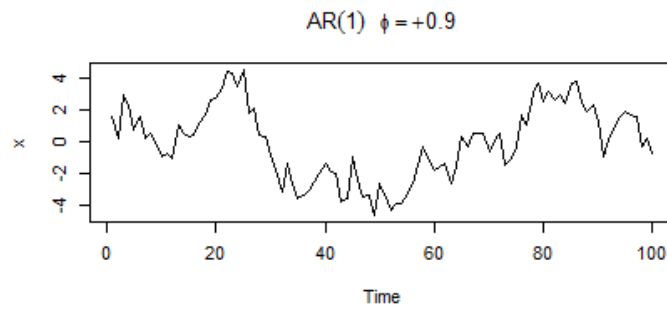
추측이 작을수록 반응이 작다..
 corr $\begin{pmatrix} Y_t & Y_{t-1} \\ Y_t & Y_{t-2} \\ & \vdots \end{pmatrix}$

ε_t : unobserved variable

filter

rejection

AutoCorrelation Function (ACF)



Properties of MA(1)

Q) Stationary?

$$Y_t = \mu + \varepsilon_t + \theta \varepsilon_{t-1}, \quad \varepsilon_t \sim WN.$$

$E(\cdot)$ on both sides. ↓

$$\begin{aligned} \textcircled{1} \quad E[Y_t] &= \mu + E[\varepsilon_t] + \theta E[\varepsilon_{t-1}] \\ &= \mu + 0 + 0 = \mu \end{aligned}$$

by definition of (V_t)

$$\textcircled{2} \quad \text{var}(Y_t) \Rightarrow E[(Y_t - \mu)^2]$$

$$Y_t = \mu + \varepsilon_t + \theta \varepsilon_{t-1}$$

$$E[Y_t - E(Y_t)]$$

μ

$$= E[(\overset{a}{\varepsilon_t} + \overset{b}{\theta \varepsilon_{t-1}})^2]$$

cov(.)

$$= E[\varepsilon_t^2] + 2\theta E[\varepsilon_t \varepsilon_{t-1}] + \theta^2 E[\varepsilon_{t-1}^2]$$

$$= \sigma_\varepsilon^2 + 0 + \theta^2 \sigma_\varepsilon^2 = \sigma_\varepsilon^2 (1 + \theta^2) = \sigma_0^2$$

Properties of MA(1)

✓

$$\begin{aligned} \text{cov}(Y_t, Y_{t-1}) &= \gamma_1 = E[(\varepsilon_t + \theta\varepsilon_{t-1})(\varepsilon_{t-1} + \theta\varepsilon_{t-2})] \\ &= E[\varepsilon_t\varepsilon_{t-1}] + \theta E[\varepsilon_t\varepsilon_{t-2}] \\ &\quad + \theta E[\varepsilon_{t-1}^2] + \theta^2 E[\varepsilon_{t-1}\varepsilon_{t-2}] \\ &= 0 + 0 + \theta\sigma_\varepsilon^2 + 0 = \underline{\theta\sigma_\varepsilon^2} \end{aligned}$$

Handwritten notes: A green arrow points from the title to the first term of the expectation. Red and purple arrows show the expansion of the product and the expectation of each term.

Furthermore,

Handwritten note: 110

$$\begin{aligned} \text{cov}(Y_t, Y_{t-2}) &= \gamma_2 = E[(\varepsilon_t + \theta\varepsilon_{t-1})(\varepsilon_{t-2} + \theta\varepsilon_{t-3})] \\ &= E[\varepsilon_t\varepsilon_{t-2}] + \theta E[\varepsilon_t\varepsilon_{t-3}] \\ &\quad + \theta E[\varepsilon_{t-1}\varepsilon_{t-2}] + \theta^2 E[\varepsilon_{t-1}\varepsilon_{t-3}] \\ &= 0 + 0 + 0 + 0 = 0 \end{aligned}$$

Handwritten notes: Blue arrows show the expansion of the product and the expectation of each term. Red circles highlight the terms that result in zero.

Similar calculation show that

Handwritten note: Blue arrow pointing down

$$\text{cov}(Y_t, Y_{t-j}) = \gamma_j = 0 \text{ for } j > 1$$

Properties of MA(1)

$$\begin{aligned} \overset{\text{corr}(\cdot)}{\rho_1} &= \overset{\text{cov}(\cdot)}{\frac{\gamma_1}{\sigma^2}} = \frac{\theta \sigma_\varepsilon^2}{\sigma_\varepsilon^2 (1 + \theta^2)} = \frac{\theta}{(1 + \theta^2)} \\ \rho_j &= \frac{\gamma_j}{\sigma^2} = 0 \text{ for } j > 1 \end{aligned}$$

Note:

$$\left(\begin{array}{l} \rho_1 = 0 \text{ if } \theta = 0 \\ \rho_1 > 0 \text{ if } \theta > 0 \\ \rho_1 < 0 \text{ if } \theta < 0 \end{array} \right)$$

Dynamic Stability Condition



- ▶ $\lim_{j \rightarrow \infty} \rho_j = 0$ is referred to as the dynamic stability condition.
- ▶ Does the MA(1) process always meet the dynamic stability condition?

Moving Average (MA) Processes

- Idea: Create a stochastic process that only exhibits two period linear time dependence

MA(2) Model

$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2},$$
$$\varepsilon_t \sim \text{iid } N(0, \sigma_\varepsilon^2)$$

Properties of MA(2)

정답
정답

$$\begin{aligned} E[Y_t] &= \mu + E[\varepsilon_t] + \theta_1 E[\varepsilon_{t-1}] + \theta_2 E[\varepsilon_{t-2}] \\ &= \mu + 0 + 0 + 0 = \mu \end{aligned}$$

HW

$$\begin{aligned} \text{var}(Y_t) &= \sigma^2 = E[(Y_t - \mu)^2] \\ &= E[(\varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2})^2] \\ &= \sigma_\varepsilon^2 + \theta_1^2 \sigma_\varepsilon^2 + \theta_2^2 \sigma_\varepsilon^2 = \sigma_\varepsilon^2 (1 + \theta_1^2 + \theta_2^2) \end{aligned}$$

Properties of MA(2)

$$\text{cov}(Y_t, Y_{t-1}) = \gamma_1 = \theta_1 \sigma_\varepsilon^2$$

$$\text{cov}(Y_t, Y_{t-2}) = \gamma_2 = \theta_2 \sigma_\varepsilon^2$$

$$\text{cov}(Y_t, Y_{t-3}) = \gamma_3 = 0$$

$$\text{cov}(Y_t, Y_{t-4}) = \gamma_4 = 0$$

$\vdots$

That is,

$$\text{cov}(Y_t, Y_{t-j}) = \gamma_j = 0 \text{ for } j > 2$$

HW



Properties of MA(2)

$$\rho_1 = \frac{\gamma_1}{\sigma^2} = \frac{\theta_1 \sigma_\varepsilon^2}{\sigma_\varepsilon^2 (1 + \theta_1^2 + \theta_2^2)} = \frac{\theta_1}{(1 + \theta_1^2 + \theta_2^2)}$$

$$\rho_2 = \frac{\gamma_2}{\sigma^2} = \frac{\theta_2 \sigma_\varepsilon^2}{\sigma_\varepsilon^2 (1 + \theta_1^2 + \theta_2^2)} = \frac{\theta_2}{(1 + \theta_1^2 + \theta_2^2)}$$

$$\rho_j = \frac{\gamma_j}{\sigma^2} = 0 \text{ for } j > 2$$

- ▶ $\lim_{j \rightarrow \infty} \rho_j = 0$ is referred to as the dynamic stability condition.
- ▶ Does the MA(2) process always meet the dynamic stability condition?

Moving Average (MA) Process of Order q

- ▶ Idea: Create a stochastic process that only exhibits linear time dependence for q periods

MA(q) Model

$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q}, \quad \varepsilon_t \sim \text{iid } N(0, \sigma_\varepsilon^2)$$

Properties of MA(q)

$$E[Y_t] = \mu$$

$$\text{var}(Y_t) = \sigma^2 = \sigma_\varepsilon^2 (1 + \theta_1^2 + \theta_2^2 + \dots + \theta_q^2)$$

$$\text{Cov}[Y_t, Y_{t-1}] = \gamma_1 \neq 0, \rho_1 \neq 0$$

$$\text{Cov}[Y_t, Y_{t-2}] = \gamma_2 \neq 0, \rho_2 \neq 0$$

$$\vdots$$

$$\text{Cov}[Y_t, Y_{t-q}] = \gamma_q \neq 0, \rho_q \neq 0$$

$$\text{Cov}[Y_t, Y_{t-q-1}] = \gamma_{q+1} = 0, \rho_{q+1} = 0$$

$$\text{Cov}[Y_t, Y_{t-q-2}] = \gamma_{q+2} = 0, \rho_{q+2} = 0$$

$$\vdots$$

Outline

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MA Model

AR Model

Wold representation and impulse response analysis

Prediction

Generalization

Auto Regressive (AR) Processes

충격이 지속성이 강하다
persistent

shock → 지속성



Idea: Create a stochastic process that exhibits multi-period geometrically decaying linear time dependence

AR(1) Model (mean-adjusted form)

$$Y_t - \mu = \phi (Y_{t-1} - \mu) + \varepsilon_t, \quad -1 < \phi < 1$$
$$\varepsilon_t \sim \text{iid } N(0, \sigma_\varepsilon^2)$$

AR(1) Model (regression form)

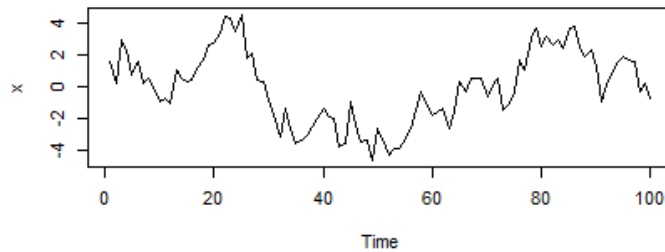
$$Y_t = c + \phi Y_{t-1} + \varepsilon_t, \quad -1 < \phi < 1$$
$$\varepsilon_t \sim \text{iid } N(0, \sigma_\varepsilon^2)$$

⇒ Partial ACF

AutoCorrelation Function (ACF)

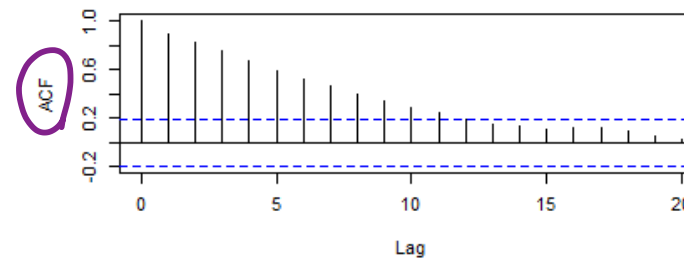
$$Y_t = \mu + \phi, Y_{t-1} + \varepsilon_t$$

AR(1) $\phi = +0.9$

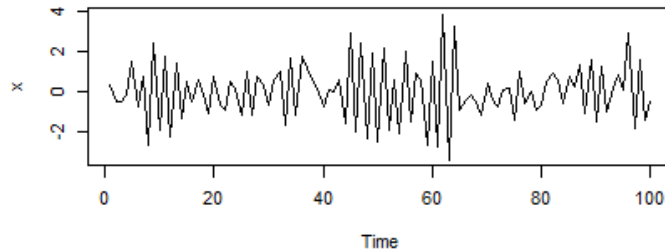


AR(1)

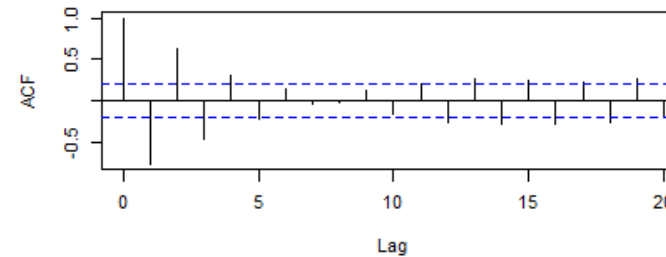
Series a1



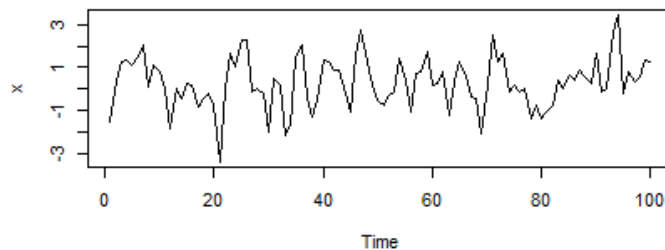
AR(1) $\phi = -0.9$



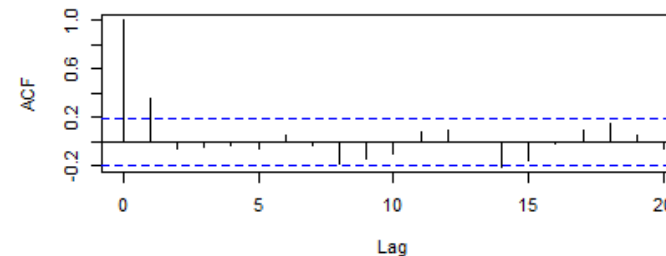
Series a2



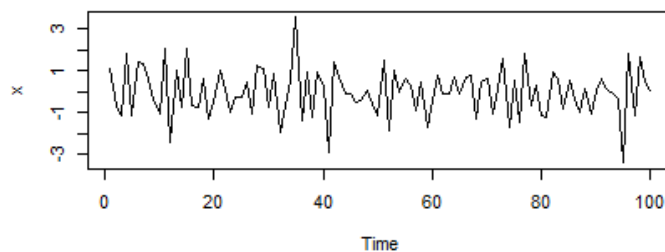
MA(1) $\theta = +0.5$



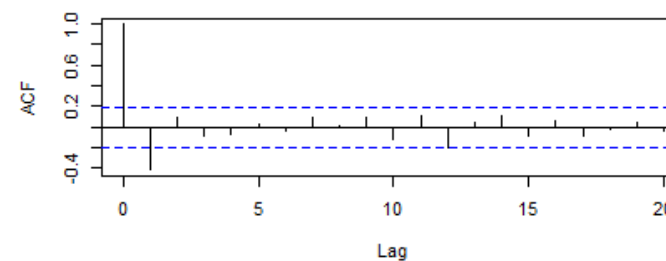
Series a3



MA(1) $\phi = -0.5$



Series a4



Properties of AR(1)

a) Stationary? ✓

Trick for calculating moments: use stationarity properties

① $E[Y_t] = E[Y_{t-j}]$ for all j ✓

② $\text{cov}(Y_t, Y_{t-j}) = \text{cov}(Y_{t-k}, Y_{t-k-j})$ for all k, j ✓

Data

o ~~~~~~~~~

- check & plot the data

- (i) Stationary? → Unit root test?

① time trend, (t)
↑
변수 → 항
② 그 외 원인
(Unit root) : 늘 증가하는 사상의
반사성



Properties of AR(1)

Mean of AR(1)

$$\begin{aligned} E[Y_t] &= c + \phi E[Y_{t-1}] + E[\varepsilon_t] \\ &= c + \phi E[Y_t] \quad // \text{ (by stationarity)} \\ \Rightarrow E[Y_t] &= \frac{c}{1-\phi} = \underline{\underline{\mu}}. \end{aligned}$$

Variance of AR(1)

$$\begin{aligned} \gamma_0 = \text{var}(Y_t) &= E[(Y_t - \mu)^2] = E[(\phi(Y_{t-1} - \mu) + \varepsilon_t)^2] \\ &= \phi^2 E[(Y_{t-1} - \mu)^2] + 2E[(Y_{t-1} - \mu)\varepsilon_t] + E[\varepsilon_t^2] \\ &= \phi^2 \gamma_0 + \sigma^2 \quad (\text{by stationarity}) \\ \Rightarrow \gamma_0 &= \frac{\sigma^2}{1-\phi^2} \quad \checkmark \end{aligned}$$

Properties of AR(1)

Trick: multiply $Y_t - \mu$ by $Y_{t-j} - \mu$ and take expectations

$$\begin{aligned}\gamma_j &= E[(Y_t - \mu)(Y_{t-j} - \mu)] \\ &= E[\phi(Y_{t-1} - \mu)(Y_{t-j} - \mu)] + E[\varepsilon_t(Y_{t-j} - \mu)] \\ &= \phi\gamma_{j-1} \text{ (by stationarity)}\end{aligned}$$

$$\Rightarrow \gamma_j = \phi^j \gamma_0 = \underbrace{\phi^j}_{\sim} \underbrace{\frac{\sigma^2}{1 - \phi^2}}_{\sim}$$

Autocorrelations:

$$\rho_j = \frac{\gamma_j}{\gamma_0} = \frac{\phi^j \gamma_0}{\gamma_0} = \phi^j$$

D.S.C.

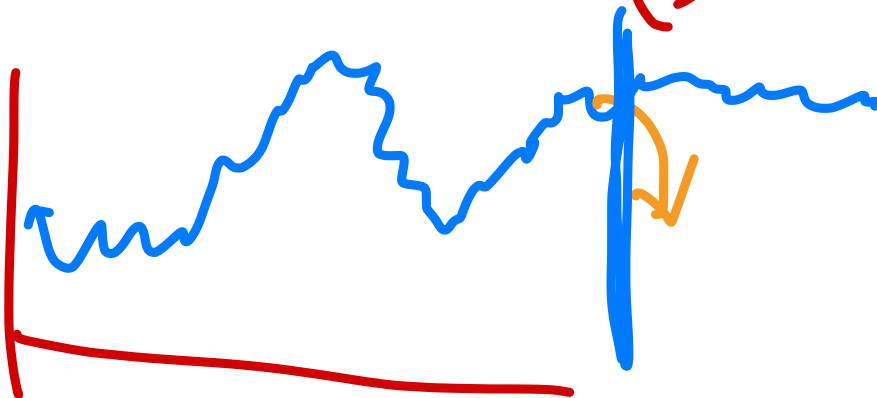
$$\lim_{j \rightarrow \infty} \rho_j = 0$$

Properties of AR(1)

$$(Y_t) \quad (Y_{t-j}) \Rightarrow \phi$$

- ▶ $\lim_{j \rightarrow \infty} \rho_j = 0$ is referred to as the dynamic stability condition.
- ▶ Does the AR(1) process always meet the dynamic stability condition?
- ▶ No. The dynamic stability condition only holds when $|\phi| < 1$.
- ▶ Basic idea of a unit root test: estimate ϕ and test for $\checkmark H_0: \phi = 1$.

$$\rho_j = \phi^j \quad (\text{AR}(1))$$



Auto Regressive (AR) Processes

• principle of
parsimony

AR(2) Model (mean-adjusted form)

$$Y_t - \mu = \phi_1 (Y_{t-1} - \mu) + \phi_2 (Y_{t-2} - \mu) + \varepsilon_t, \varepsilon_t \sim \text{iid } N(0, \sigma_\varepsilon^2)$$

Handwritten annotations: A '+' sign above the first '+' and a circled '+' with an arrow pointing to the second '+'.

AR(2) Model (regression form)

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t, \varepsilon_t \sim \text{iid } N(0, \sigma_\varepsilon^2)$$

Properties of AR(2)

Stationary $\Rightarrow$?

Mean of AR(2)

$$\begin{aligned} E[Y_t] &= c + \phi_1 E[Y_{t-1}] + \phi_2 E[Y_{t-2}] + E[\varepsilon_t] \\ &= c + \phi_1 E[Y_t] + \phi_2 E[Y_t] \\ &\Rightarrow E[Y_t] = \frac{c}{1 - \phi_1 - \phi_2} = \mu \end{aligned}$$

cbv stationary 718)

Variance of AR(1)

$$\begin{aligned} \gamma_0 = \text{var}(Y_t) &= E[(Y_t - \mu)^2] = E[(\phi(Y_{t-1} - \mu) + \varepsilon_t)^2] \\ &= \phi^2 E[(Y_{t-1} - \mu)^2] + 2E[(Y_{t-1} - \mu)\varepsilon_t] + E[\varepsilon_t^2] \\ &= \phi^2 \gamma_0 + \sigma^2 \quad (\text{by stationarity}) \\ &\Rightarrow \gamma_0 = \frac{\sigma^2}{1 - \phi^2} \end{aligned}$$

$\phi_1 \quad \phi_2$

✓

Properties of AR(2) . . . AR(2), "Matrix"

Variance of AR(2): We need to use a matrix form of AR(2).

AR(1)

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t$$

or

State vector

$$\begin{bmatrix} Y_t \\ Y_{t-1} \end{bmatrix} = \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} Y_{t-1} \\ Y_{t-2} \end{bmatrix} + \begin{bmatrix} \varepsilon_t \\ 0 \end{bmatrix}$$

Matrix

$$\begin{matrix} \zeta_t \\ (2 \times 1) \end{matrix} = \begin{matrix} \mathbf{F} \\ (2 \times 2) \end{matrix} \begin{matrix} \zeta_{t-1} \\ (2 \times 1) \end{matrix} + \begin{matrix} \mathbf{v}_t \\ (2 \times 1) \end{matrix}, \mathbf{v}_t \sim N(0, \mathbf{R})$$

$$Y_{t-1} = Y_{t-1}$$

ζ_t

$$\text{where } \mathbf{R} = \begin{bmatrix} \sigma_\varepsilon^2 & 0 \\ 0 & 0 \end{bmatrix}.$$

Properties of AR(2)

推定. $Var [\xi_t] = \mathbf{F} Var [\xi_t] \mathbf{F}^\top + Var [\mathbf{v}_t]$

$$\boxed{Vec} (Var [\xi_t]) = Vec (\mathbf{F} Var [\xi_t] \mathbf{F}^\top) + Vec (Var [\mathbf{v}_t])$$

$$Vec (Var [\xi_t]) = \boxed{\mathbf{F} \otimes \mathbf{F}} Vec (Var [\xi_t]) + Vec (\mathbf{R})$$

$$(\mathbf{I}_4 - \mathbf{F} \otimes \mathbf{F}) Vec (Var [\xi_t]) = Vec (\mathbf{R})$$

$$Vec (Var [\xi_t]) = (\mathbf{I}_4 - \mathbf{F} \otimes \mathbf{F})^{-1} Vec (\mathbf{R})$$

krona product : element by element multiplication

- ▶ The first element of matrix $(\mathbf{I}_4 - \mathbf{F} \otimes \mathbf{F})^{-1} Vec (\mathbf{R})$ is $E[Y_t]$.

Properties of AR(2)

$$\rho_j = ?$$

$$\frac{\partial \gamma_j}{\partial \epsilon_t} = ?$$

cov

Trick: multiply $Y_t - \mu$ by $Y_{t-j} - \mu$ and take expectations

$$\begin{aligned}\gamma_j &= E[(Y_t - \mu)(Y_{t-j} - \mu)] \\ &= E[\phi_1(\underline{Y_{t-1} - \mu})(\underline{Y_{t-j} - \mu})] + E[\phi_2(Y_{t-2} - \mu)(Y_{t-j} - \mu)] \\ &\quad + E[\epsilon_t(Y_{t-j} - \mu)] \\ &= \phi_1 \underline{\gamma_{j-1}} + \phi_2 \underline{\gamma_{j-2}} \text{ (by stationarity)}\end{aligned}$$

Properties of AR(2)

- How do we check stationarity? We need to check if $\lim_{j \rightarrow \infty} \rho_j = 0$ holds.

Dynamic equation for autocorrelations:

$$\rho_j = \phi_1 \rho_{j-1} + \phi_2 \rho_{j-2}$$

AR(1) ✓

or

$$\begin{bmatrix} \rho_j \\ \rho_{j-1} \end{bmatrix} = \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \rho_{j-1} \\ \rho_{j-2} \end{bmatrix}$$

$$\underset{(2 \times 1)}{\boldsymbol{\rho}_j} = \underset{(2 \times 2)}{\mathbf{F}} \boldsymbol{\rho}_{j-1}$$

Finding Eigenvalues

λ is an eigenvalue of $\mathbf{F}$ and $\mathbf{x}$ is an eigenvector if

$$\mathbf{F}\mathbf{x} = \lambda\mathbf{x} \Rightarrow (\mathbf{F} - \lambda\mathbf{I}_p)\mathbf{x} = 0$$

$$\Rightarrow \mathbf{F} - \lambda\mathbf{I}_p \text{ is singular} \Rightarrow \det(\mathbf{F} - \lambda\mathbf{I}_p) = 0$$

Example: $AR(2)$

$$\begin{aligned}\det(\mathbf{F} - \lambda\mathbf{I}_2) &= \det\left(\begin{pmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{pmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right) \\ &= \det\begin{pmatrix} \phi_1 - \lambda & \phi_2 \\ 1 & -\lambda \end{pmatrix} \\ &= \lambda^2 - \phi_1\lambda - \phi_2\end{aligned}$$

The eigenvalues of $\mathbf{F}$ solve the characteristic equation

$$\lambda^2 - \phi_1\lambda - \phi_2 = 0$$

Finding Eigenvalues

The eigenvalues of **F** solve the characteristic equation

$$\lambda^2 - \phi_1\lambda - \phi_2 = 0$$

Using the quadratic equation, the roots satisfy

$$\lambda_i = \frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{2}, i = 1, 2$$

Finding Eigenvalues

To see why $|\lambda_i| < 1$ implies $\lim_{j \rightarrow \infty} \boldsymbol{\rho}_j = \mathbf{0}$ consider the $AR(2)$ with real-valued eigenvalues. By the spectral decomposition theorem

$$\mathbf{F} = \mathbf{T}\mathbf{\Lambda}\mathbf{T}^{-1}, \mathbf{T}^{-1} = \mathbf{T}'$$

$$\mathbf{\Lambda} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}, \mathbf{T} = \begin{bmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{bmatrix},$$

$$\mathbf{T}^{-1} = \begin{bmatrix} t^{11} & t^{12} \\ t^{21} & t^{22} \end{bmatrix}$$

Finding Eigenvalues

$$\rho_j = F \cdot \rho_{j-1}$$

Substitution

$$\vdots$$
$$\rho_j = F^j \rho_0$$

...

Then

$$\textcircled{F^j} = \overset{7F^2}{(\mathbf{T}\mathbf{\Lambda}\mathbf{T}^{-1})} \times \cdots \times (\mathbf{T}\mathbf{\Lambda}\mathbf{T}^{-1})$$
$$= \mathbf{T}\mathbf{\Lambda}^j\mathbf{T}^{-1}$$

and

$$\lim_{j \rightarrow \infty} \mathbf{F}^j = \mathbf{T} \lim_{j \rightarrow \infty} \mathbf{\Lambda}^j \mathbf{T}^{-1} = 0$$

provided $|\lambda_1| < 1$ and $|\lambda_2| < 1$.

$\Rightarrow$ D.S.C

► $\rho_j = \mathbf{F}\rho_{j-1}$ means $\rho_j = \mathbf{F}^j\rho_0$.

► $\lim_{j \rightarrow \infty} \rho_j = \lim_{j \rightarrow \infty} \mathbf{F}^j\rho_0 = 0$

► $\lim_{j \rightarrow \infty} \rho_j = 0$ provided $|\lambda_1| < 1$ and $|\lambda_2| < 1$.



Auto Regressive (AR) Processes

AR(p) Model (mean-adjusted form)

$$Y_t - \mu = \phi_1 (Y_{t-1} - \mu) + \phi_2 (Y_{t-2} - \mu) + \dots + \phi_p (Y_{t-p} - \mu) + \varepsilon_t$$

AR(p) Model (regression form)

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \varepsilon_t$$

Properties of AR(p)

- ▶ Can you find the mean of AR(p) process?
- ▶ Can you find the variance of AR(p) process?
- ▶ Can you find the covariance of AR(p) process?
- ▶ Can you derive the stationarity condition for AR(p)?

Auto Regressive Moving Average (ARMA) Processes

ARMA(p,q) Model (mean-adjusted form)

$$Y_t - \mu = \phi_1 (Y_{t-1} - \mu) + \phi_2 (Y_{t-2} - \mu) + \dots + \phi_p (Y_{t-p} - \mu) + \varepsilon_t \\ + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q}$$

ARMA(p,q) Model (regression form)

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \varepsilon_t \\ + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q}$$

Properties of AR(p)

- ▶ Can you find the mean of ARMA(p,q) process?
- ▶ Can you find the variance of ARMA(p,q) process?
- ▶ Can you find the covariance of ARMA(p,q) process?
- ▶ Can you derive the stationarity condition for ARMA(p,q)?

Outline

Motivation

MA Model

AR Model

Wold representation and impulse response analysis

Prediction

Generalization

Wold's Decomposition Theorem

Any covariance stationary time series $\{Y_t\}$ can be represented in the form

$$Y_t = \mu + \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}, \quad \varepsilon_t \sim i.i.d. (0, \sigma^2)$$

$$\psi_0 = 1, \quad \sum_{j=0}^{\infty} \psi_j^2 < \infty$$

- Parsimonious parametric models like ARMA models approximate the Wold form of stationary time series.

Impulse response function: AR(1)

AR(1) Model: (ignore an intercept term for simplicity)

$$Y_t = \phi Y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2)$$

- Solution by recursive substitution:

$$\begin{aligned} Y_t &= \phi^{t+1} Y_{-1} + \phi^t \varepsilon_0 + \cdots + \phi \varepsilon_{t-1} + \varepsilon_t \\ &= \phi^{t+1} Y_{-1} + \sum_{i=0}^t \phi^i \varepsilon_{t-i} \\ &= \phi^{t+1} Y_{-1} + \sum_{i=0}^t \psi_i \varepsilon_{t-i}, \quad \psi_i = \phi^i \end{aligned}$$

- Alternatively, solving forward j periods from time t :

$$\begin{aligned} Y_{t+j} &= \phi^{j+1} Y_{t-1} + \phi^j \varepsilon_t + \cdots + \phi \varepsilon_{t+j-1} + \varepsilon_{t+j} \\ &= \phi^{j+1} Y_{t-1} + \sum_{i=0}^j \psi_i \varepsilon_{t+j-i} \end{aligned}$$

Impulse response function: AR(1)

$\Rightarrow$ AR(p)

State vector

AR(1) Model: (ignore an intercept term for simplicity)

$\vdots$
VAR(1)

$$Y_t = \phi Y_{t-1} + \varepsilon_t, \varepsilon_t \sim N(0, \sigma^2)$$

► Dynamic Multiplier:

$$\frac{dY_t}{d\varepsilon_{t-j}} = \frac{dY_j}{d\varepsilon_0} = \frac{dY_{t+j}}{d\varepsilon_t} = \phi^j = \psi_j$$

► Interpretation is ... Given everything constant, how a change in ε_t will affect y_{t+j} . (Causal inference)

► Impulse Response Function (IRF)

Plot ψ_j against j

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Prediction: AR(1)

AR(1) Model: (ignore an intercept term for simplicity)

$$Y_t = \phi Y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma^2)$$

- Prediction is based on:

$$Y_{t+j} = \phi^j Y_t + \phi^{j-1} \varepsilon_{t+1} + \phi^{j-2} \varepsilon_{t+2} + \dots + \phi \varepsilon_{t+j-1} + \varepsilon_{t+j}$$

- The conditional expectation and variance of Y_{t+j} given I_t is

$$E[Y_{t+j} | I_t] = \phi^j Y_t$$

$$\text{Var}[Y_{t+j} | I_t] = \phi^{2(j-1)} \sigma^2 + \phi^{2(j-2)} \sigma^2 + \dots + \sigma^2$$

where the current information contains all observations up to time t , $I_t = \{Y_1, Y_2, \dots, Y_t\}$.

$$Y_{t+j} - E[Y_{t+j} | I_t] = \text{prediction error.}$$

$$Y_{t+1} = \phi Y_t + \varepsilon_{t+1}$$
$$Y_{t+2} = \phi Y_{t+1} + \varepsilon_{t+2}$$

Outline

Motivation

MA Model

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Wold representation and impulse response analysis

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Impulse response function: ARMA(2,2)

ARMA(2,2) Model: (ignore an intercept term for simplicity)

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$$

$$\varepsilon_t \sim N(0, \sigma^2)$$

► The matrix representation of ARMA(2,2).

$$\begin{bmatrix} Y_t \\ Y_{t-1} \\ \varepsilon_t \\ \varepsilon_{t-1} \end{bmatrix} = \begin{bmatrix} \phi_1 & \phi_2 & \theta_1 & \theta_2 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} Y_{t-1} \\ Y_{t-2} \\ \varepsilon_{t-1} \\ \varepsilon_{t-2} \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} \varepsilon_t$$

$$\underset{(4 \times 1)}{\xi_t} = \underset{(4 \times 4)}{\mathbf{F}} \xi_{t-1} + \underset{(4 \times 1)}{\mathbf{v}_t}, \mathbf{v}_t = \underset{(1 \times 4)}{\mathbf{R}} \varepsilon_t$$

$$\text{where } \mathbf{R} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}.$$

Impulse response function: ARMA(2,2)

- Solving forward j periods from time t :

$$\zeta_{t+j} = \mathbf{F}\zeta_{t+j-1} + \mathbf{v}_{t+j}$$

$$= \mathbf{F}^2\zeta_{t+j-2} + \mathbf{F}\mathbf{v}_{t+j-1} + \mathbf{v}_{t+j}$$

⋮

$$= \mathbf{F}^j\zeta_t + \mathbf{F}^{j-1}\mathbf{v}_{t+1} + \cdots + \mathbf{F}\mathbf{v}_{t+j-1} + \mathbf{v}_{t+j}$$

$$= \mathbf{F}^{j+1}\zeta_{t-1} + \mathbf{F}^j\mathbf{v}_t + \mathbf{F}^{j-1}\mathbf{v}_{t+1} + \cdots + \mathbf{F}\mathbf{v}_{t+j-1} + \mathbf{v}_{t+j}$$

$$= \mathbf{F}^{j+1}\zeta_{t-1} + \sum_{i=0}^j \mathbf{F}^i \mathbf{v}_{t+j-i}$$

$$= \mathbf{F}^{j+1}\zeta_{t-1} + \sum_{i=0}^j \mathbf{F}^i \mathbf{R} \varepsilon_{t+j-i}$$

$$\zeta_t = \mathbf{F} \zeta_{t-1} + \mathbf{v}_t$$

$$\mathbf{v}_t = \mathbf{R} \cdot \varepsilon_t$$

3-4

Impulse response function: ARMA(2,2)

OLS.

NLF

ARMA(2,2) Model: (ignore an intercept term for simplicity)

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$$

$$\varepsilon_t \sim N(0, \sigma^2)$$

► Dynamic Multiplier:

$$\frac{dY_t}{d\varepsilon_{t-j}} = \frac{dY_j}{d\varepsilon_0} = \frac{dY_{t+j}}{d\varepsilon_t} = \text{first element of } \mathbf{F}^j \mathbf{R} = \psi_j$$

► Interpretation is ... Given everything constant, how a change in ε_t will affect y_{t+j} . (Causal inference)

► Impulse Response Function (IRF)

Plot ψ_j against j

Prediction: ARMA(2,2)

ARMA(2,2) Model: (ignore an intercept term for simplicity)

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$$

$$\varepsilon_t \sim N(0, \sigma^2)$$

$$\mathbf{F}^j \boldsymbol{\xi}_t = \mathbb{E}[\mathbf{F}^j \boldsymbol{\xi}_t | \mathcal{I}_t]$$

- Prediction is based on:

$$\begin{aligned} \boldsymbol{\zeta}_{t+j} &= \mathbf{F}^j \boldsymbol{\zeta}_t + \mathbf{F}^{j-1} \mathbf{v}_{t+1} + \cdots + \mathbf{F} \mathbf{v}_{t+j-1} + \mathbf{v}_{t+j} \\ &= \mathbf{F}^j \boldsymbol{\zeta}_t + \mathbf{F}^{j-1} \mathbf{R} \varepsilon_{t+1} + \cdots + \mathbf{F} \mathbf{R} \varepsilon_{t+j-1} + \mathbf{R} \varepsilon_{t+j} \end{aligned}$$

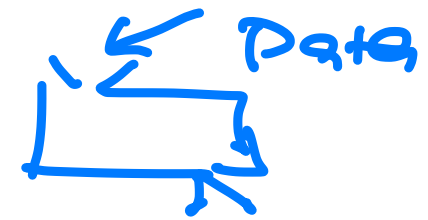
- The conditional expectation and variance of Y_{t+j} given $\mathcal{I}_t$ is

$$\begin{aligned} \mathbb{E}[Y_{t+j} | \mathcal{I}_t] &= \text{the first element of } \mathbf{F}^j \mathbb{E}[\boldsymbol{\zeta}_t | \mathcal{I}_t] \\ \text{Var}[Y_{t+j} | \mathcal{I}_t] &= \text{the first by first element of} \end{aligned}$$

$$\text{Var}[\mathbf{F}^{j-1} \mathbf{R} \varepsilon_{t+1} + \cdots + \mathbf{F} \mathbf{R} \varepsilon_{t+j-1} + \mathbf{R} \varepsilon_{t+j}]$$

where the current information contains all observations up to time t , $\mathcal{I}_t = \{Y_1, Y_2, \dots, Y_t\}$.

Prediction: ARMA(2,2)



ARMA(2,2) Model: (ignore an intercept term for simplicity)

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$$

$$\varepsilon_t \sim N(0, \sigma^2)$$

- ▶ For a general ARMA model, it is difficult to compute the conditional mean and variance of Y_{t+j} given information up to time t because ζ_t contains unobserved error terms, ε_t and ε_{t-1} .
- ▶ Therefore, we typically rely on the Kalman filtering algorithm to compute the conditional moments for prediction. We will cover this later in class.
- ▶ We can perform impulse response analysis for a general ARMA(p,q) model without the Kalman filter because only thing we need to know is the coefficient matrix **F**.

Statistical inference and prediction with ARMA(p,q)

For a general ARMA model, we can do impulse response analysis and prediction.

- Step 1: Use MLE (we will learn this estimation method) to estimate all model parameters.
- Step 2: Construct the matrix representation of your ARMA model.
- Step 3: Given the coefficient matrix $\mathbf{F}$, perform impulse response analysis and prediction.

The End. Next, we will learn how to estimate ARMA models. After that we will learn unit root tests to statistically check if a time series variable is stationary or non-stationary.